

Adem Toumi

Artificial Intelligence & Data Science Engineer

+213 795 016 471 | Biskra, Algeria | addz1606@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

AI & Data Science Engineer trained at the Higher School of Computer Science (ESI-SBA), Algeria. Solid background across Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Large Language Models, Software Engineering, and AI Application Development. Focused on applying theoretical foundations to design reliable intelligent systems and practical software applications.

EDUCATION

Higher School of Computer Science (ESI-SBA)

State Engineer's Degree & M.Sc. in Artificial Intelligence and Data Science

Sidi Bel Abbès, Algeria

Sep. 2021 – Jul. 2026

- **Curriculum:** Specialized in machine learning, deep learning, computer vision, natural language processing, distributed systems, big data architectures, and software engineering.
- **Thesis & Graduation Project (PFE):** *Log-Based Anomaly Detection in Distributed Systems* (Joint Engineering & Master's Thesis under supervision of Dr. Sidi Mohammed Benslimane & Dr. Mohammed Bekkouche).

TECHNICAL SKILLS

AI & Data Science:	Python, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, OpenCV, XGBoost, FastAPI
Data Engineering:	SQL, PostgreSQL, Apache Spark, Apache Kafka, MLflow
Programming Languages:	Python, Java, Dart, JavaScript, TypeScript
Web & Mobile:	React, Next.js, Flutter, Node.js, HTML, Tailwind CSS, Android (Java)
Tools & Infrastructure:	Git, Docker, Linux, HuggingFace Spaces

RESEARCH & INDUSTRY EXPERIENCE

Research Intern – Object Detection & Tracking for UAV Systems

Feb. 2025 – Jun. 2025

Electronic Systems Research and Development Unit (URD Laboratory), Sidi Bel Abbès, Algeria

- Fine-tuned object detection models including YOLOv11s on aerial imagery datasets.
- Integrated DeepSort and CSRT multi-object tracking algorithms across sequential video frames.
- Evaluated edge-device real-time inference performance on Raspberry Pi 5 with Intel Movidius NCS 2 hardware acceleration.
- Developed a Tkinter-based Ground Control Station (GCS) featuring live video stream decoding and telemetry over custom UDP protocols.

Intern – Database & Network Administration

Sep. 2024 – Present

Sonatrach, Direction Maintenance, Biskra, Algeria

- Performed database administration tasks including SQL query optimization and schema maintenance.
- Supported network administration, system monitoring, and automated data backup scripts.

SELECTED PROJECTS

Artificial Intelligence & Data Science Projects

KnowledgeHub AI – Production RAG Platform (Flagship AI Project) *Python, FastAPI, React, FAISS, BM25, Gemini, Docker*

- Designed and developed a production-ready Retrieval-Augmented Generation (RAG) platform for intelligent document understanding and question answering.
- Engineered a Hybrid Search pipeline combining dense vector retrieval (FAISS) with sparse lexical search (BM25) and Cross-Encoder reranking to maximize context precision.
- Integrated Explainable AI (XAI) for transparent source attribution, Google Gemini LLM synthesis, React dashboard, and Docker containerization.

Log-Based Anomaly Detection in Distributed Systems (PFE & Thesis)

Python, PyTorch, Kafka, FastAPI, Streamlit

- Evaluated 11 machine learning, deep learning, and transformer models (including SVM, Random Forest, BiLSTM, Autoencoders, and LogBERT) across distributed system log datasets under strict temporal data splitting.
- Built a real-time streaming pipeline using Apache Kafka for data ingestion, cloud-deployed FastAPI services for inference, PostgreSQL for storage, and an interactive Streamlit monitoring dashboard.
- Integrated post-hoc Explainable AI (SHAP and LIME) to provide local and global interpretability for log-level anomaly detection.

Object Detection and Tracking for UAV Applications (Academic Project)

Python, PyTorch, OpenCV, YOLO

- Fine-tuned deep learning object detection models (YOLOv11s) on aerial imagery datasets for target identification under drone surveillance conditions.
- Integrated multi-object tracking algorithms (DeepSort and CSRT) across sequential video streams to ensure continuous object tracking under camera motion and scale variation.
- Evaluated edge-device real-time inference on Raspberry Pi 5 accelerated by Intel Movidius NCS 2, and built a Tkinter Ground Control Station (GCS) with live UDP telemetry.

Real-Time Monitoring and Anomaly Detection (Deep Learning Project)

Python, PyTorch, Prometheus, Grafana

- Designed a time-series monitoring pipeline for system CPU/GPU metrics using Prometheus and Grafana.
- Built and trained a hybrid VAE-LSTM model in PyTorch to detect abnormal operational behavior and send live notifications.

Multi-Class Malicious URL Detection (Machine Learning Project)

Python, Scikit-Learn, XGBoost, TensorFlow

- Developed a machine learning classifier to identify malicious web URLs using character n-grams and structural URL features.
- Compared multiple classification models (Random Forest, XGBoost, LSTM) to deliver fast, content-independent web security analysis.

Paraphrase Detection System (NLP Project)

Python, PyTorch, spaCy, NLTK

- Designed a Siamese neural network with shared weights to detect semantic sentence similarity across sentence pairs.
- Utilized Bi-LSTM layers and multi-head attention mechanisms to capture contextual semantic relationships.

Movie Recommendation System (Mini Project)

Python, Scikit-Learn, MLflow, Streamlit

- Implemented a recommender system using collaborative filtering and association rule mining (Apriori, FP-Growth).
- Tracked experiment parameters using MLflow and deployed an interactive Streamlit user interface.

Software Engineering Projects

HajjPortal – Multilingual Pilgrimage Platform

TypeScript, Next.js 14, React, Socket.io, TanStack Query

- Developed a full-stack, trilingual (Arabic, English, French) web platform with Next.js 14 App Router, complete RTL/LTR layout support, and Zod validation.
- Integrated a real-time administrative control dashboard using Socket.io and TanStack Query state synchronization.

KidLink – School Communication Platform

Dart, Flutter, Firebase, Provider

- Built a cross-platform mobile application using Flutter and Provider to streamline parent-teacher communication in primary schools.
- Integrated Firebase Authentication, Cloud Firestore for offline-first data synchronization, and Firebase Cloud Messaging (FCM) for push notifications.

XAI Log Anomaly Detection Showcase

React 18, Vite 5, Framer Motion, Tailwind CSS

- Developed an interactive single-page web application using React 18, Vite, and Tailwind CSS to display PFE benchmark findings.
- Implemented interactive data visualizations using Recharts and explainable AI galleries.

LANGUAGES

Arabic | French | English